

TROUBLESHOOTING GUIDE

Diagnostic & Incident Resolution Manual — Vishal Gold

Project	Vishal Gold — B2B/B2C Jewelry App
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1. Native Build & Environment Conflicts

1.1 Gradle Sync & Execution Failures

Problem: Native Android builds fail with 'Could not resolve all files' or 'Gradle sync failed'.

Resolution: Run 'cd android && ./gradlew clean'. If the issue persists, delete the '.gradle' folder in the user home directory and ensure JAVA_HOME is pointing to a compatible OpenJDK (v17+).

1.2 MultiDex Constraint

Problem: Build fails on method reference count (65k limit).

Resolution: App is MultiDex ready. Confirm 'minSdkVersion 21' is set in build.gradle. Avoid adding oversized plugins unnecessarily.

2. State Management (Provider) Failures

2.1 ProviderNotFoundException

Problem: App crashes with a red error screen stating the provider could not be found.

Resolution: Ensure the widget accessing the provider is a descendant of MultiProvider in main.dart. Avoid using GlobalKeys to bypass the widget tree context.

2.2 LateInitializationError

Symptoms: 'Field xxx has not been initialized'.

Resolution: This usually occurs when LocalStorageService.init() is skipped in main(). Verify the 'await' keyword is present before the app runs.

3. Firebase Connectivity & Permissions

3.1 [cloud_firestore/permission-denied]

Problem: Firestore reads/writes fail silently or with an exception.

Resolution: Audit Firestore Security Rules. Ensure the user's UID exists in the /users/ collection with the 'wholesaler' or 'retailer' flag correctly set.

3.2 Cloud Storage 403 (Unauthorized)

Problem: Admin cannot upload product images.

Resolution: Check individual image size (>10MB may timeout). Verify admin role permissions in the Storage bucket rules.

4. Authentication & OTP Reliability

4.1 SMS Quota/Delivery Issues

Symptoms: OTP is not received after 60 seconds.

Resolution: 1. Confirm device phone number contains (+91). 2. Check Firebase Console for 'SMS Quota Exceeded'. 3. Use pre-defined Test Numbers for development to avoid hitting the live SMS quota.

4.2 'Too Many Requests' Lockout

Problem: Firebase blocks the device temporarily.

Resolution: This is a server-side safety trigger. Wait and use the 60-second cooldown timer correctly. Do not clear app storage to bypass, as this creates a new session and potentially extends the lockout.

5. Security Barriers & Lockouts

5.1 Screenshot Prevention

Observation: User reports they cannot screenshot the cart or profile.

status: THIS IS BY DESIGN. FLAG_SECURE is active on sensitive pages to protect PII. It is not an error.

5.2 ADB Log Exposure

Problem: Sensitive data appears in 'adb logcat'.

Resolution: All debug logging must be wrapped in 'kDebugMode' or 'kProfileMode' guards and removed from the Release branch.

6. Diagnostic Command Reference

Action	Command
Health Check	flutter doctor -v
State Inspection	flutter pub global run devtools
Native Logs	adb logcat -s flutter
Clean Build	flutter clean && flutter pub get
Release Test	flutter run --release

7. Emergency Recovery Procedure

In cases of systemic instability or inconsistent local state:

1. Clear App Data: Settings -> Apps -> Home -> Clear Storage.
2. Force Sync: Long-press the logo to verify the admin entry is removed (it should be).
3. Re-Authenticate: Perform a full login cycle to re-establish secure tokens.